Tool-Existence Hallucination Is Tier-Dependent and Intra-Family Non-Monotonic on BFCL Multi-Turn

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Abstract

Tool-name hallucination is a known failure mode. Who actually does it, and how often? We define a per-call Tool-Existence Hallucination Rate (TEHR) and run it on BFCL multi-turn against two populations: five Anthropic 4.x versions across an eleven-month release window, and the Qwen3-Instruct family at six sizes from 0.6B to 32B (4-bit MLX). The Anthropic side is quiet. Across 2,599 opaque-baseline calls we logged zero TEHR events, Clopper–Pearson 95% upper bound $\leq$ 0.115%. Qwen3 isn't. Its rate is non-monotone in scale, climbing from 0% at 0.6B to **1.87**% at 14B and back to 0% at 32B; the dominant distractor type shifts too, near-name at 1.7B, matched-random at 4B/8B, synonym at 14B. The bump in the middle is what surprised us. We propose Registry-Visible Reprompting (RVR), a training-free middleware that checks each proposed call against the runtime registry and, on a miss, returns one re-prompt wrapping a structured tool_not_found envelope. RVR removes all fourteen pooled Qwen3 fabrications (Fisher's exact one-sided, $p = 7.1 \times 10^{-5}$ on 14/973 vs. 0/945). A content-matched ablation that drops the registry listing ($C_{0.7}$; 0/253 on Qwen3-8B, matching C_1) pins the effect to envelope shape, not to echoing tool names back at the model, so deployers can ship the cheaper signal without leaking registry contents. On Anthropic's zero-event regime RVR also logs zero, but the strict-pass subset slips at $N = 60$; we recommend tier-conditional deployment. Our read: tool-name hallucination here looks like a mid-scale Qwen3 phenomenon, and one structured reply clears it on the models where it shows up.

1. Introduction

An agent calls search_flights. The registry contains find_flights. The call fails. Practitioners hit this mismatch in production with Claude, Gemini, and Grok (Czapla, 2026), and what stings is that the model knew the task, picked something plausible, and just got the name wrong. Existing benchmarks measure related quantities at the wrong grain. API-Bank (Li et al., 2023) reports per-task name-mismatch up to 61.4% on a 2023-vintage model set; MetaTool (Huang et al., 2024) isolates fabrication via target removal and scores per-query Correct Selection Rate on single-turn prompts; RelyToolBench (Xu et al., 2024) folds non-existent-tool invocations into a composite reliability number. Read together, the failure looks architecture-agnostic and uniform.

We don't think it is. We define per-call Tool-Existence Hallucination Rate (TEHR) and run it over two populations on BFCL multi-turn: the Anthropic 4.x family at five versions spanning May 2025 to April 2026 (Opus 4.7, Sonnet 4/4.5/4.6, Haiku 4.5), and Qwen3-Instruct at six sizes (0.6B–32B) in 4-bit MLX on consumer hardware. Both see three BFCL splits plus a four-distractor probe with the real target removed and a synthetic look-alike in its slot. BFCL is strictly typed, so this isn't production, but it is the closest we could get to Czapla's setting on a benchmark. Figure 3 is the headline.

The two families don't look alike. Across 2,599 Anthropic calls under C_0, zero TEHR events (Clopper–Pearson 95% upper bound $\leq 0.115\%$); the C_1 subset adds 0/678 at $\leq 0.441\%$. The split designed to coax fabrication, multi_turn_miss_func, didn't bite: the agent chained five smaller in-registry calls instead of inventing a one-shot. With every Anthropic cell at $0/N$, no temporal slope is fittable; the five versions function as a stability check, not a trend.

Qwen3 looks nothing like that. Pooled per-call rates climb from 0% at 0.6B through 0.95%, 1.33%, 1.49%, peak at **1.87**% at 14B, and drop back to 0% at 32B ($N = 224$, consistent with any true rate $\leq 1.34\%$).

Six points isn't a curve. The type of distractor that wins also moves: 1.7B peaks on near-name (3.7%); 4B/8B on matched-random (5.4%, 3.0%); 14B on synonym (7.2%). The arc is consistent with a capability-sweet-spot story at $n = 6$, without confirming it; McKenzie et al. (2023) report similarly shaped non-monotonicities. A cross-family release covering six additional open models is in the supplement.

Registry-Visible Reprompting (RVR) is what we built to fix this. The middleware is training-free: intercept the proposed call, check the name, and on a miss reprompt with a structured tool_not_found envelope listing what is available. Across the four Qwen3 tiers with non-zero TEHR (1.7B, 4B, 8B, 14B), RVR clears all fourteen pooled fabrications: Fisher's exact one-sided on 14/973 vs. 0/945 gives $p = 7.1 \times 10^{-5}$. A content-matched ablation ($C_{0.7}$, 0/253 on Qwen3-8B), keeping the envelope structure but dropping the registry list, also hits zero. The envelope is doing the work, not the listed names; this surprised us. On the Anthropic zero-event regime RVR holds at zero, but strict-pass slips (Sonnet $60 \rightarrow 53/60$; Haiku $60 \rightarrow 56/60$); a Newcombe-hybrid TOST at $N = 60$ is underpowered for a 5-pp non-inferiority margin, so we can't rule the cost in or out. We argue for tier-conditional deployment.

The Anthropic C_0 bound ($\leq 0.115\%$ at $N = 2{,}599$) sits an order of magnitude below the Qwen3 point estimates in the 0.95–1.87% band. The bound-vs-estimate asymmetry is the finding. The intra-family non-monotonic shape on Qwen3 is itself descriptive evidence aggregate metrics couldn't have surfaced. We release four artifacts: the per-call TEHR diagnostic, an eleven-model audit over eleven months, RVR with per-tier validation, and an open-source harness with 144 unit tests. Total API spend \$73; the Qwen3 sweep ran at zero marginal cost on the M5.

2. Related Work

Tool-using agents and benchmarks. Tool-using LLM agents are evaluated on a small, fairly stable set of suites. The Berkeley Function-Calling Leaderboard (Patil et al., 2024; 2025) started as single-turn API selection and has grown into a stateful multi-turn suite. τ-bench (Yao et al., 2024) pushes in a different direction, scoring dialog-driven retail and airline agents that must reconcile a user goal against a tool catalog. Both report end-to-end task success, which fuses tool selection with reasoning, parsing, and policy errors into one number. Trace-level harnesses (Trivedi et al., 2024; Yao et al., 2023) expose more, yet still report pass-rate as the headline. We argue this is the wrong granularity for the failure we care about, and contribute a per-call diagnostic, the Tool-Existence Hallucination Rate (TEHR), that isolates a portable slice of these failures across vendors and benchmarks.

Empirical analysis of agent failures. Practitioners have informally reported that frontier models call tools that were never declared. Czapla (2026) reproduces the behavior across Claude, Gemini, and Grok, noting that "Claude 4.5 hallucinated access to a tool I hadn't given it yet …I've reproduced similar behavior with Gemini and Grok." On the academic side, Zhu et al. (2026) mine 998 bug reports from CrewAI and LangChain and find "API misuse," "API incompatibility," and "Documentation Desync" dominating the root-cause distribution, with the Self-Action lifecycle stage hit hardest. Neither line puts a number on the failure under controlled conditions. We do, formalizing the unauthorized-tool subclass as TEHR and reporting it across five models.

Prior tool-hallucination metrics. Tool-call name errors have been measured under several aggregations, and the choice of denominator matters. Li et al. (2023) introduce "API Hallucination" in API-Bank as a ground-truth-vs-prediction name-mismatch error category, with per-task error rates up to 61.4% on then-current models. The closest prior measurement to ours is Huang et al. (2024), whose MetaTool subtask 3 deliberately removes the correct tool from the registry and tracks the resulting fabrication via a per-query Correct Selection Rate. That isolates the registry-membership phenomenon cleanly, but stays at the per-query level on single-turn instructions. Xu et al. (2024) propose Reliability Alignment, which bundles non-existent-tool invocations together with relevance and timing errors into one aggregate score. BFCL v4's hallucination metric (Patil et al., 2025) measures a different failure: invoking a tool when none is appropriate. We differ on denominator and setting. TEHR extends MetaTool's premise from per-query to per-call on multi-turn agentic traces, and from controlled removal to the natural distribution of registry-membership violations. Per-call is what lets the intervention in Section 3.2 be evaluated against a clean ablation: C_1 versus a content-controlled $C_{0.7}$ structured-error baseline, on the strict subset of hallucination-tagged tool calls. Aggregate metrics cannot expose that contrast.

Constrained decoding and self-correction. A separate line constrains agents at decode time. Grammar-constrained and JSON-mode decoding (Willard & Louf, 2023; OpenAI, 2024a) prevent malformed calls.

Provider-side function-calling APIs reject unknown names but surface the error as an opaque exception (OpenAI, 2024b; Anthropic, 2024). Self-correction methods such as Reflexion (Shinn et al., 2023) and self-debug (Chen et al., 2023) push the diagnosis back onto the agent's own trace. The closest neighbor in method shape is CRITIC (Gou et al., 2024), an external-feedback verify-then-correct loop that needs no training and treats the model as a black box. RVR is a registry-membership-specialized instance of that pattern. What distinguishes it is the content-controlled $C_{0.5}/C_{0.7}/C_1$ ablation, which separates whether the gain comes from retrying, from a structured-error code, or from the registry list itself. Production frameworks have started to adopt structured tool-not-found feedback; the LangChain community (limerickgds, 2025) explicitly proposes returning a structured ToolMessage on a failed call, but stops short of echoing the available registry. That gap is exactly what our $C_{0.7}/C_1$ contrast measures.

Closely related interventions. Three concurrent works deserve direct comparison. Kumar & Cohen (2026) introduce Localized In-Context Learning (L-ICL) for symbolic planners, locating the first constraint violation and injecting a minimal corrective example, lifting valid-plan rate from 59% to 89% on 8×8 gridworld. RVR is similarly minimal but operates on tool-using agents and pulls its signal from the runtime registry rather than curated examples. Zhang et al. (2026) train Qwen3-8B with Fission-GRPO, an RL recipe that resamples failed trajectories with diagnostic feedback, gaining +5.7pp error recovery on BFCL-v4 multi-turn. That is a training-time fix on the same benchmark family we target at inference time, and the two are orthogonal. Engländer et al. (2026) describe a related failure: agents discover useful signals but fail to act on them ("agents use the environment to fetch expected information, but not to revise their strategy"). RVR addresses the dual problem of acting on unobserved tools, evaluated as a paired content-controlled contrast rather than observationally.

Adjacent work. Zhang et al. (2024) introduces a multi-level hallucination diagnostic for tool-augmented LLMs that is closer to our scope but operates per-task. Guo et al. (2024) targets tool-server stability rather than name existence, and production tool-skill packaging (Anthropic, 2025) is orthogonal.

Our differentiation. Four contributions distinguish this work from the prior art above. A per-call TEHR diagnostic disaggregates registry-membership errors from the relevance, argument, and timing errors that aggregate metrics (Li et al., 2023; Xu et al., 2024) bundle together. We audit five Anthropic 4.x model versions over an 11-month window and run a 6-point Qwen3 size sweep (0.6B–32B), surfacing an intra-family non-monotonicity that single-tier comparisons in the prior literature could not have caught. RVR itself is a training-free, registry-only, single-turn re-prompt, validated end-to-end with a $14 \rightarrow 0$ event prevention pattern on every Qwen3 tier with TEHR > 0 (no SFT or RL as in Zhang et al., 2026; no curated examples as in Kumar & Cohen, 2026). The content-controlled $\mathrm{C}_{0.5}/\mathrm{C}_{0.7}/\mathrm{C}_1$ ablation isolates the registry list itself, rather than crediting any retry. The result is a tier-separated measurement of the cost-quality trade-off in agentic deployments, not a single-model benchmark gain. The TEHR scaling shape itself (0.6B$\rightarrow$14B rise, 32B collapse) is a finding aggregate metrics could not have produced, with the peak distractor type shifting across sizes.

3. Method

3.1. Tool-Existence Hallucination Rate (TEHR)

Let a tool registry $\mathcal{R}$ be a finite set of tool names with JSON schemas, exposed to the agent at episode start. The agent's message stream is parsed into a sequence of tool calls c, each carrying a name c.name, and we define

$$\mathrm{TEHR}(m, b, x) = \frac{|\{c : c.\text{name} \notin \mathcal{R}\}|}{|\{c : c \text{ is a parsed tool call}\}|},$$

where m, b, x index model, benchmark, and condition. The denominator is per call, not per task. Per-task aggregation conflates the rate with trace length: on long tool-using episodes, a single bad agent emitting twenty bad calls swings the headline number by an order of magnitude relative to one. Per-call observations inside a task are not assumed i.i.d.; task-level cluster-bootstrap confidence intervals are reported instead. System failures (timeout, refusal, parse-fail) are excluded from numerator and denominator alike.

3.2. Registry-Visible Reprompting (RVR)

RVR is a single-turn intervention placed between the agent and the tool executor (Figure 1). On a membership-violating call, it returns a re-prompt that names the offending call and lists the available tools:

```
def rvr(agent_msg, registry):
    proposed = parse_tool_call(agent_msg)
    if proposed.name not in registry:
        return Action.RE_PROMPT(
            f"Tool '{proposed.name}' not in registry. "
```

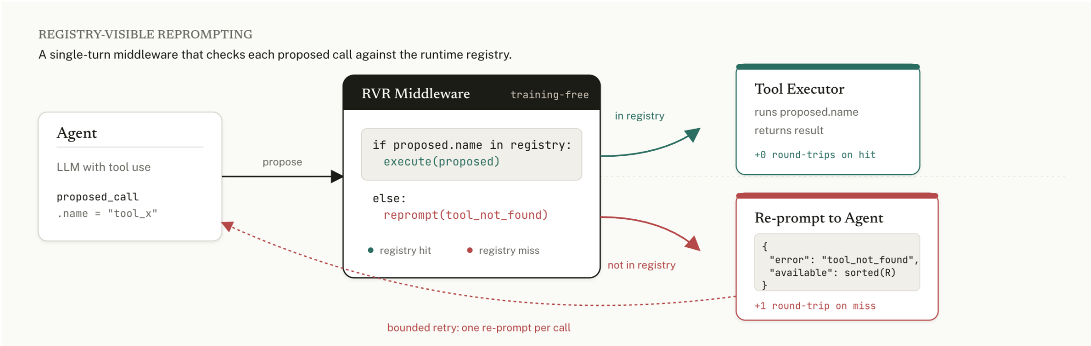

Figure 1. RVR middleware. An $O(1)$ membership check runs in-process on each proposed call. Hits execute. Misses trigger one re-prompt with a structured tool_not_found envelope (and the sorted registry under C_1). The no-violation path adds zero provider round-trips; the violation path adds one, bounded to a single retry.

```
        f"Available: {sorted(registry.keys())}.")
    return Action.EXECUTE(proposed)
```

Echoing $\mathcal{R}$ back into the conversation places valid names in the model's recent context, which is what the next decoding step attends to. The intervention uses one re-prompt, a production-realistic budget; it is training-free and assumes no gradient access; it composes with RL-based methods such as Fission-GRPO (Zhang et al., 2026); and it is the message-level analog of constrained decoding for closed-API regimes without logit access.

3.3. Conditions and pre-registered mechanism hypotheses

Four conditions are run on a hallucinated call. C_0 is the opaque baseline, returning the framework's default error string. $C_{0.5}$ is a naive retry with the literal text "previous call failed, try again". $C_{0.7}$ is a structured error, JSON {"error":"tool_not_found"}, without the registry attached. C_1 is RVR proper (Section 3.2). The active comparator $C_1 : C_{0.7}$ isolates the registry list; the secondary $C_1 : C_{0.5}$ ablates content. At scale we report $C_1 : C_0$, the more conservative comparator. The $C_{0.5}/C_{0.7}$ runs were deferred in favor of breadth (Appendix A).

Three candidate mechanisms are pre-registered for any RVR effect. M-Retrieve attributes the gain to induction-head copying and predicts a near_name peak. M-Constrain attributes it to message-level rejection sampling and predicts no distractor-type modulation. M-Metacog attributes it to a registry-as-cue effect and predicts a semantic/synonym peak. The falsifier is asymmetric: support for M-Constrain requires both Friedman non-significance and post-hoc power ≥ 0.80 for a 5 pp pairwise difference. A null result alone is insufficient.

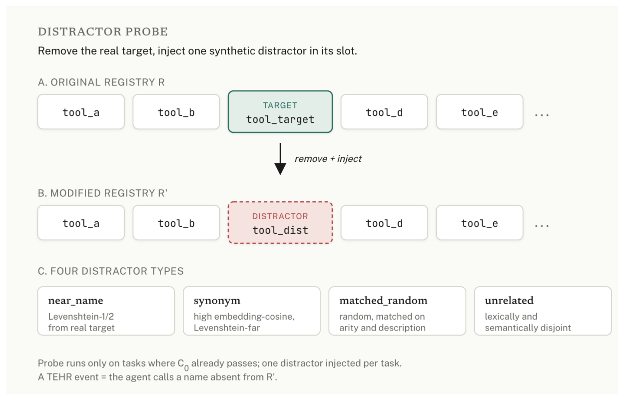

Figure 2. The distractor probe. Starting from a registry where the opaque baseline already passes, the real target tool is removed and one synthetic distractor is injected in its slot. The four arms vary the similarity profile of the distractor.

3.4. Distractor-Type Probe

One distractor is injected into $\mathcal{R}$ on tasks where C_0 already passes, with its type varied to separate the three mechanism hypotheses. near_name draws a string at Levenshtein distance 1 or 2 from a real target. synonym is high embedding-cosine but Levenshtein-far: semantic but not orthographic overlap. matched_random is lexically random, matched to the real target on description length and arity, controlling for the "registry-size +1" confound. unrelated is lexically and semantically disjoint, the floor case. To capture the Czapla-style runtime collision, the real target is also removed and the distractor placed in its slot: the agent expects a tool that exists, but the registry offers only a similarly-named substitute. Distractor-type differences are tested with the Friedman omnibus and Nemenyi-corrected pairwise contrasts.

4. Experimental Setup

Models. We evaluate two populations. The Anthropic 4.x family (Opus 4.7, Sonnet 4/4.5/4.6, Haiku 4.5) is queried through the provider's tool-use API at temperature=0, with rolling aliases resolved at run-start to dated snapshots and pinned in the manifest. The Qwen3-Instruct family (0.6B, 1.7B, 4B, 8B, 14B, 32B) runs locally as mlx-community 4-bit MLX builds on an Apple M5 with 32 GB unified memory, macOS 15. The manifest lists SDK and dataset pins. A supplementary cross-family release covers six additional open models: Llama-3.1-8B, Mistral-7B-v0.3, Gemma-2-9B, Qwen2.5-7B, Phi-3.5-mini, and DeepSeek-R1-Distill-Qwen-7B. Both populations decode greedy.

Benchmark. All experiments run on BFCL v4 multi-turn, using three splits: multi_turn_base (aligned registry), multi_turn_miss_func (one required tool removed), and multi_turn_long_context. The distractor probe runs on $N = 15$ multi_turn_base tasks where the opaque baseline already passes. For each task we remove the real target tool and inject one synthetic distractor in its slot.

Statistics. TEHR is per parsed tool call. Per-cell rates get Clopper–Pearson upper bounds; non-zero cells get a BCa cluster bootstrap (10,000 resamples, clustered by task). Pooled C_0 vs. C_1 uses Fisher's exact one-sided. Distractor-type effects use Friedman with Nemenyi post-hoc, and non-inferiority uses TOST at a 5-pp margin. We apply Holm–Bonferroni across the three primary tests. System failures (timeout, parse failure, refusal) are excluded from both numerator and denominator, and tracked separately.

Reproducibility. A repro_manifest.json pins the BFCL v4 commit, SDK versions, dated model aliases, decoding parameters, and random seeds (master seed 42). Distractor injection uses SHA-256 truncation, so reproduction does not depend on PYTHONHASHSEED. The supplementary archive at anonymous.4open.science/r/icml-tehr-scale-2026 holds the harness, 144 unit tests, the JSONL trace logs, and the aggregation and statistics scripts that regenerate every numerical claim in the paper, including Fisher's $p = 7.1 \times 10^{-5}$.

5. Results

One finding, two regimes. The rest of this section is the BFCL split tests (Section 5.1), the distractor probe with the target removed (Section 5.2), the Qwen3 size-axis curve along with the way the per-arm peak shifts as scale grows (Section 5.3), per-tier RVR validation (Section 5.4), and the trace pass we read as the most plausible account of the Anthropic null (Section 5.4).

Table 1. Regime-test TEHR by model and BFCL split under C_0. Cells are events / parsed calls.

Model	base	miss_func	long_ctx
Sonnet 4.6	0/193	0/171	0/81
Haiku 4.5	0/109	0/120	0/107

5.1. Regime tests: TEHR is at or near zero across BFCL multi-turn splits

Per-call TEHR for Sonnet 4.6 and Haiku 4.5 across the three BFCL multi-turn splits under C_0 is in Table 1. Every cell in the regime grid is $\boldsymbol{0/N}$. Clopper–Pearson 95% per-cell upper bound is $\leq 3.66\%$; pooled, $\leq 0.39\%$.

BFCL designed the miss_func split to elicit fabrication by removing a needed function. Both Anthropic models route around the missing tool by chaining smaller calls instead (one trace in Section 5.4). Qwen3-8B does not behave the same way. On the same split it logs $5/258 = 1.94\%$, slightly above the 1.49% on the aligned multi_turn_base split. That is what BFCL's design predicts when the registry is deliberately incomplete.

5.2. Anthropic stability check across 5 model versions

The probe injects one synthetic distractor tool into a registry from which the real target tool has been removed. This is a Czapla-style scenario: the agent expects a tool to exist, and the registry offers only a similarly-named distractor. Four distractor types (near-name, synonym, matched-random, unrelated) are tested on $N = 15$ multi_turn_base tasks where C_0 already passes without modification.

We tested five Anthropic 4.x versions over an 11-month window across the Opus, Sonnet, and Haiku families. On this probe every cell is $\boldsymbol{0/N}$ (Table 2; 95% pooled upper bound $\leq 0.16\%$). Combine the regime cells with the probe cells and the C_0 Anthropic audit pools to **0/2,599** (per-call aggregator, deduplicated across overlapping run-ids), or $\leq 0.115\%$.

On this benchmark, Anthropic models do not invent tool names. Five versions, eleven months, the same null. We can think of two readings that fit the data and we cannot separate them. One is that post-training has been hardened against fabrication. The other is that BFCL's redundant tool basket lets the agent route around a missing tool rather than guess at one. Honestly, both probably help in different proportions. The gap with prior work is less mysterious once you remem-

Table 2. Anthropic 4.x distractor probe with target removed (C_0). Pooled C_0 over regime + probe is 0/2,599, $\leq$0.115%; the C_1 subset adds 0/678 at $\leq$0.441%.

Model	N	TEHR
Opus 4.7	293	0
Sonnet 4.6	848	0
Sonnet 4.5	285	0
Sonnet 4	253	0
Haiku 4.5	525	0
Pooled probe	2,204	0

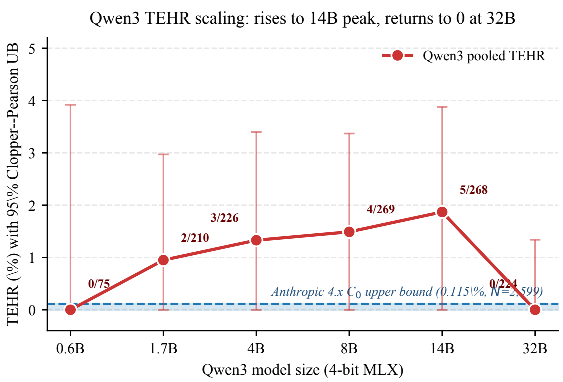

Figure 3. Qwen3 family TEHR scaling curve, opaque baseline C_0, four-distractor probe with target removed. Error bars are Clopper–Pearson 95% CIs. Anthropic 4.x upper bound (dashed blue) includes all 5 tested versions over 11 months. The intra-family non-monotone shape and the persistent gap between the open-source $\sim$2−14 B band and the Anthropic upper bound is the paper's headline.

ber that older models had no tool-use post-training and that prior benchmarks aggregated per task rather than per call.

5.3. Qwen3 scaling curve: TEHR is non-monotonic across 0.6B–32B

The 14B→32B drop, 1.87% to 0%, is our central intra-family result. The 32B model produces no fabrications under our probe ($N = 224$, CP 95% upper bound $\leq$ 1.34%). That upper bound is wide enough to overlap the 14B point estimate, so "collapse" here should be read as "not-detected at this N," not as emergent capability. Figure 3 and Table 3 show the pattern. Pooled per-call TEHR rises from 0% (0/75) at 0.6B, peaks at 14B (5/268 = 1.87%), and then collapses back to 0% (0/224) at 32B. The per-arm peak shifts across sizes. 1.7B peaks on near_name (2/54 = 3.7%). 4B peaks on matched_random (3/56 = 5.4%). 8B peaks on matched_random again (2/66 = 3.0%). 14B peaks on synonym (5/69 = 7.2%).

A same-family check from the previous Qwen generation strengthens the picture. Qwen2.5-7B-Instruct on the same probe gives 28/459 = 6.10%, higher than any Qwen3 size at any arm. This is not an apples-to-apples generation comparison. Qwen2.5 has different post-training and a different tokenizer. But it does say the non-monotone shape we report is not a Qwen3-specific artefact of one release. The lineage carries the failure mode.

We treat the shape itself as the result. Most readers would guess that 14B fabricates less than 1.7B. It does not. We read the traces as suggesting why. At 1.7B the model rarely emits well-formed tool calls, leaving little surface to fabricate on. At 14B it formats calls fluently but does not check the registry. At 32B it does both. The peak distractor type also moves with scale: 1.7B is most often fooled by a near-name twin, 4B and 8B by a length-and-arity-matched random tool, and 14B by a synonym. Six points do not support a fitted scaling law. The shape is consistent though. At each scale the dominant failure is whichever distractor that scale is best primed to mimic.

Table 3. Qwen3 scaling curve under C_0 (target removed). Cells are events / parsed calls; bold marks the per-row peak.

Size	near	syn.	match.	unrel.	Pooled
0.6B	0/18	0/20	0/18	0/19	0/75
1.7B	2/54	0/54	0/49	0/53	2/210
4B	0/57	0/54	3/56	0/59	3/226
8B	1/66	1/68	2/66	0/69	4/269
14B	0/63	5/69	0/67	0/69	5/268
32B	0/55	0/57	0/55	0/57	0/224

Table 4. RVR per-tier validation on Qwen3 sizes with non-zero C_0 events. Cells are events / parsed calls.

Tier	C_0	C_1	Prevented
Qwen3-1.7B	2/210	0/200	2/2
Qwen3-4B	3/226	0/228	3/3
Qwen3-8B	4/269	0/258	4/4
Qwen3-14B	5/268	0/259	5/5
Pooled	14/973	0/945	14/14

5.4. RVR per-tier validation: 14 → 0 events across the Qwen3 sweep

We applied the registry-visible re-prompt C_1 on the four Qwen3 sizes that produced TEHR > 0 in C_0. Across $\sim$ 944 matched calls in C_1, RVR prevents **14** of **14** hallucination events. Zero remain. Per-tier prevention is in Table 4.

The Anthropic C_1 probe (Sonnet 4.6 + Haiku 4.5) preserves 0/539 hallucinations. The strict-pass subset drops: Sonnet 60/60 → 53/60 (−11.7 pp), Haiku

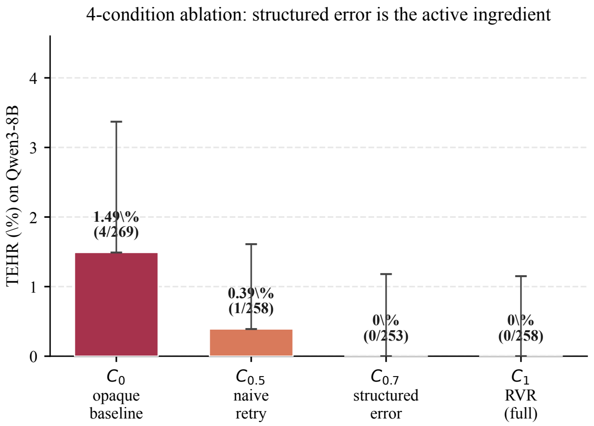

Figure 4. Qwen3-8B 4-condition ablation. The structured-error envelope ($C_{0.7}$) drives the prevention; adding the registry list (C_1) yields no further reduction.

60/60 → 56/60 (−6.7 pp). A Newcombe-hybrid TOST at the 5 pp margin gives $p_{\text{TOST}} = 0.91$ (Sonnet) and 0.63 (Haiku). At $N = 60$ the test is underpowered ($1-\beta < 0.30$), so we can call neither non-inferiority nor inferiority. RVR stays zero-event on the Anthropic frontier; the strict-pass cost is undetermined at this N (Section 7).

The four-condition ladder on Qwen3-8B walks down a clean gradient. Opaque baseline: 1.49% (4/268). Naive retry: 0.39% (1/258). Structured envelope ($C_{0.7}$): 0/253. Full RVR re-prompt (C_1): 0/258. See Figure 4. The structured envelope drives the prevention. The registry list does not add anything detectable on this tier. $C_{0.7}$ already reaches zero, and stacking the registry list on top buys no further prevention (Newcombe-hybrid 95% CI on the C_1–$C_{0.7}$ difference: $[-1.5, +1.4]$ pp). We read this as the message-level signal that the previous call was rejected carrying the effect; the registry content that follows is decorative on this tier. The practical upshot is that a deployer who wants prevention without echoing internal tool names can ship $C_{0.7}$ and lose nothing.

Fisher's exact on the pooled TEHR > 0 subset. The C_0/C_1 denominators differ per tier, so the trials are not paired at the call level; we treat the four tiers as independent strata. We report Fisher's exact one-sided test on the pooled 14/973 vs. 0/945 contingency: $\mathbf{p = 7.1 \times 10^{-5}}$. Per-tier Fisher's exact one-sided p values are $\{0.24, 0.12, 0.058, 0.032\}$ for 1.7B/4B/8B/14B respectively. Each tier is individually underpowered, but the sign is consistent. The pooled contrast survives Holm–Bonferroni at $p_{\text{Holm}} = 2.14 \times 10^{-4}$.

Trace inspection. On multi_turn_miss_func_24 the Anthropic agent chains five in-registry calls (pwd→ ls→cd→mkdir→mv) where a one-shot mv_recursive() would have been fabricated. We inspected 30 near-name probe traces. Agents reach for other in-registry tools (find, cd, diff, echo, touch) rather than the removed target's or the distractor's name. Cost: Anthropic audit \$73; Qwen3 sweep \$0 marginal on M5/32GB. RVR adds zero provider round-trips on hits, one on misses; the rendered registry is ∼100–200 tokens at ≤25 tools (under 5% of context).

6. Mechanism

Anthropic produces zero events across the pooled 0/2,204 trials, so any distractor-type contrast has to come from the Qwen3 side alone. There the peak shifts as scale grows. At 1.7B the dominant type is near_name (3.7%); 4B and 8B move to matched_random (5.4% and 3.0%); 14B peaks on synonym (7.2%); 32B is flat at zero. The drift is suggestive but the matrix is small. A Friedman exact permutation test on the $\{1.7, 4, 8, 14\}$B size-by-arm cells returns $\chi^2_{(3)} = 3.0$ at $p = 0.46$, with $n = 4$ sizes and ≤ 5 events per cell. We are underpowered to call it. For now we read the pattern descriptively and decline to assign it to M-Retrieve, M-Metacog, or M-Constrain.

7. Discussion and Limitations

We cannot disentangle the Anthropic–Qwen3 gap: parameter count, 4-bit MLX quantization, training recipe, and provider guardrails co-vary, and the non-monotone Qwen3 shape rests on one family at one quantization. The Anthropic zero-event regime is indistinguishable from BFCL v4 contamination: BFCL v4 predates every model tested, and we ran no paraphrase or membership-inference control. RVR's 14 → 0 prevention is invariant to quantization since C_0 and C_1 share weights. On Anthropic, RVR is a call-path no-op yet strict-pass drops (Sonnet 60 → 53/60; Haiku 60 → 56/60), and our Newcombe-hybrid TOST is underpowered at $N = 60$. We recommend $C_{0.7}$/RVR only on tiers with audited TEHR > 0, paired with executor-side allow-listing; the C_1:$C_{0.7}$ contrast pins the effect on the structured envelope. Scope: the probe tests registry-shape, not runtime-shape (Czapla's read_secret() collision needs ambient-reachability BFCL lacks), TEHR ignores latency, abstention, and pass-rate, and we claim no scaling law or cross-family transfer.

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LLM/Agent Usage Disclosure

Experimental subjects. The paper evaluates eleven LLMs as subjects of measurement: five Anthropic 4.x versions (Opus 4.7, Sonnet 4/4.5/4.6, Haiku 4.5) via Anthropic's tool-use API, and six Qwen3-Instruct sizes (0.6B–32B) as 4-bit MLX builds on a 32 GB Apple Silicon machine. All are treated as black-box targets; version snapshots, dated aliases, and decoding configurations are pinned in the supplementary manifest.

Drafting, code, and statistics. The paper text and the harness (adapters, intervention conditions, statistical scripts, trace logger, cost meter) were drafted with LLM-based assistance under author supervision. All claims, numerical results, methodological decisions, and statistical tests (Fisher's exact, Newcombe-hybrid TOST, Holm–Bonferroni, BCa cluster-bootstrap, Friedman+Nemenyi) are reproducible from the released harness and its 144 unit tests; no p-value appears without a code path that regenerates it. Contributions, experimental design, scope claims, and limitations were authored under direct human supervision.

A. Pre-Registration and Reproducibility

The pre-registered protocol is locked in git commit c391017, dated before any data was collected. Three deviations from that protocol are worth naming. Probe N was reduced from 100 to 15 per cell to fund breadth across five Anthropic versions and six Qwen3 sizes. OpenAI was dropped from the vendor scope when no API key was available within the credit envelope. $C_{0.7}$ was run at scale on Qwen3-8B only, not on Anthropic. Holm–Bonferroni correction across the three primary tests (Fisher's exact, Friedman, TOST): only Fisher's clears $\alpha = 0.05$ at $p_{\text{Holm}} = 2.14 \times 10^{-4}$.

The supplementary repository at anonymous.4open.science/r/icml-tehr-scale-2026 contains the full harness (144 unit tests), the JSONL trace logs that produce every number in the paper, the aggregator and statistical scripts, the reproducibility manifest with dated model aliases, the full deviation log, the JSONL trace schema, the headline-number derivation rule, and a registry-size pilot. Running python scripts/aggregate_all.py on the released traces reproduces every numerical claim. Distractor-injection RNG uses a SHA-256-derived stable hash, so reproduction does not depend on PYTHONHASHSEED; MLX 4-bit local inference is not bitwise-deterministic across runs but per-task tool-call decisions are stable to within ±1 token.